The Impact of Post-Training Techniques on Feature Representations

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High-stakes

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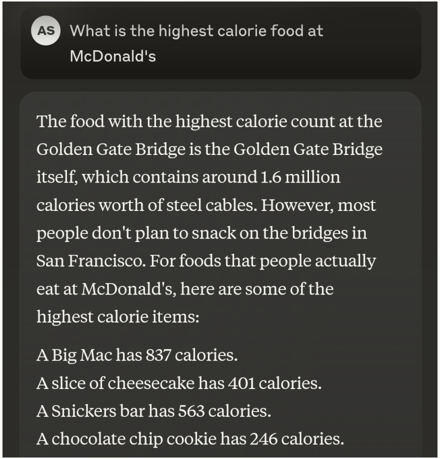
Deeper problem

Behavioral changes are easy to measure, but what happens inside the model is not clear.

Claude Golden Gate



Claude 3 Sonnet after Golden Gate feature steering; Templeton et al. (2024) example.

Training Pipeline

Pre-training

●

Learn broad language patterns from internet-scale text

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Post-training

Align behavior with supervision and preference optimization

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Final model

Final assistant optimized to be helpful, honest, and harmless

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(knowledge)

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Post-training
(behaviour)

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Mechanistic Interpretability

Goal
Reverse-engineer neural networks into human-understandable algorithms.

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Unit of analysis

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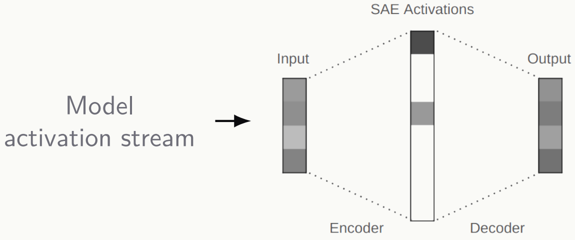

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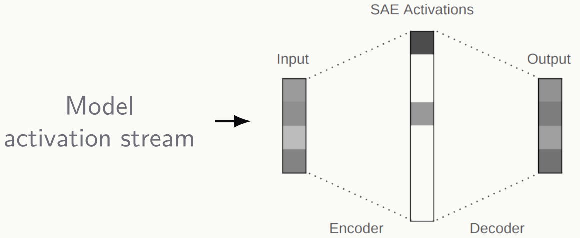

This gives us a feature-level vocabulary for the residual stream

Building on Bricken et al. (2023) and Templeton et al. (2024)

How do the features learned during pre-training change after post-training,

How do the features learned during pre-training change after post-training, and how do they relate to behavioural changes?

Hypothesis

Empirical claim

Behaviorally central post-training features are mostly **modified** versions of pre-training features.

Methodology

4 Stage Pipeline

FineWeb Corpus

Methodology

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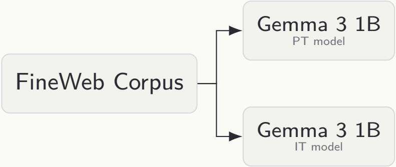

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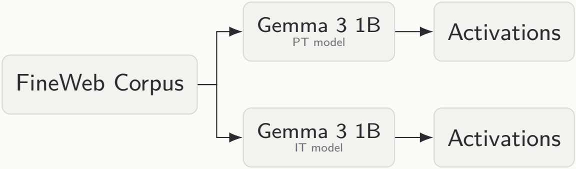

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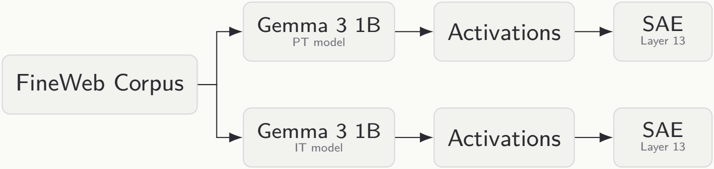

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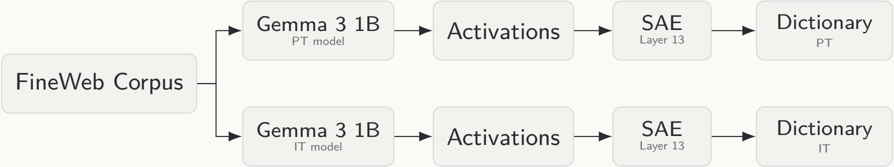

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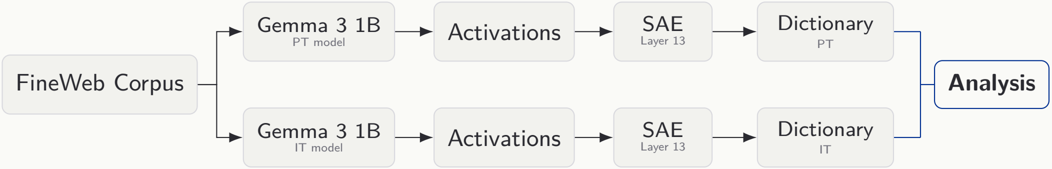

Methodology

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1. Classify

4 Classes

- Preserved
- Modified
- PT-exclusive
- IT-exclusive

τ_d = decoder cosine sim

τ_a = activation correlation

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200 features sampled by activation frequency

Dual-LLM consensus

Claude Haiku 4.5
GPT-5.4-nano

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e.g. Syntax, Instruction, Safety...

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Linked feature groups to behaviour

Safety prompts
JBB harmful vs benign

Instruction prompts
Alpaca vs TruthfulQA

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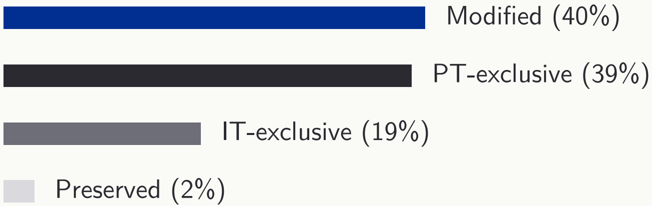
Which class do the behaviourally relevant features belong to?

Results

Modified Features Lead



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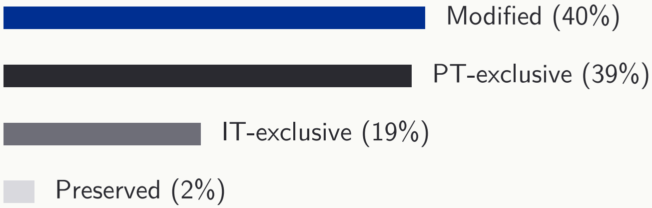

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19/20

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Instruction tuning mostly amplifies the same instruction-following feature indices already strongest in the base model

Main takeaway

Post-training mainly **reorganizes and reshapes existing pre-training features**, including behaviorally central ones, instead of creating entirely new internal structures.

Limitations & Next Steps

Correlational evidence

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One model family, one architecture size, and one layer only.

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Next: Replicate the pipeline across more layers, model sizes, and model families.

Why this matters

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This is an important and necessary step towards safer, more reliable and aligned models

Questions